1. Interpreting the coagulopathy

Example: PT 35 s, aPTT 55 s, fibrinogen 0.8, platelets 45, D-dimer > 20,000

- Consumption + dilution + liver synthetic failure
- Low fibrinogen + very high D-dimer + oozing from lines → DIC and/or hyperfibrinolysis on background of cirrhosis

Rebalanced haemostasis in cirrhosis

- ↓ pro- and anticoagulant factors (protein C, AT); ↑ factor VIII and vWF
- INR does **not** predict bleeding; global tests (TEG/ROTEM) better

Key point: Do not treat the INR; treat bleeding guided by global tests.

2. Liver disease vs DIC vs hyperfibrinolysis

Liver disease

- Normal/↑ factor VIII, low factor V and VII, stable picture, splenomegaly-related thrombocytopenia

DIC

- ISTH score ≥ 5 (platelets, D-dimer/FDP, PT, fibrinogen); trigger (sepsis, shock, trauma)
- ↓ factor VIII (consumed), falling platelets and fibrinogen over time, schistocytes

Hyperfibrinolysis

- Very low fibrinogen, very high D-dimer, oozing; ROTEM: **ML > 15%** in EXTEM, corrected by APTEM; euglobulin lysis time short

Key point: Factor VIII separates them: high in liver disease, low in DIC.

3. Goal-directed haemostatic management

General

- Control the source (endoscopy, TIPS, surgery), warm, Ca2+ > 1.1, pH > 7.2
- Restrictive transfusion Hb 70 g/L in variceal bleeding (Villanueva 2013 ↓ mortality)

Targets

- Fibrinogen > 1.5 g/L (> 2 in obstetric): cryoprecipitate 10 units or concentrate 3–4 g
- Platelets > 50 $\times 10^9$/L if bleeding (> 100 for neurosurgery)
- FFP: only if ROTEM CT prolonged or active bleeding with ↑ PT; volume overload ↑ portal pressure
- Tranexamic acid for hyperfibrinolysis (HALT-IT: no benefit in GI bleed and ↑ VTE)

Key point: Give the specific product for the specific defect.

4. Variceal bleed bundle

Steps

- Terlipressin/octreotide, antibiotics (ceftriaxone), endoscopic band ligation < 12 h
- Airway protection; restrictive transfusion
- Pre-emptive TIPS (< 72 h) in Child C 10–13 or B with active bleed (Baveno VII)
- Balloon tamponade/esophageal stent as bridge

Key point: Vasoactive drug + antibiotics + early banding + consider early TIPS.

5. Recombinant factor VIIa

Evidence

- Variceal bleeding (Bosch 2004, 2008): no benefit overall
- Trauma (CONTROL): ↓ transfusion, no mortality benefit
- Off-label use ↑ arterial thrombosis (Levi 2010 NEJM)

If used

- Last resort after surgical control and correction of fibrinogen, platelets, Ca, pH, temperature

Key point: Not recommended routinely: thrombosis risk without survival benefit.

6. Viscoelastic tests (TEG/ROTEM)

Reading ROTEM

- CT (clotting time) ↑ → factors → FFP/PCC
- A5/A10 FIBTEM < 8–10 mm → fibrinogen
- EXTEM low A5 with normal FIBTEM → platelets
- ML > 15% → antifibrinolytic
- HEPTEM vs INTEM → heparin effect

Evidence

- Cirrhosis procedures: TEG-guided ↓ products without more bleeding (De Pietri 2016)
- Cardiac surgery and trauma: ↓ transfusion

Limitations

- Insensitive to antiplatelet drugs, vWF, hypothermia (run at 37 °C)

Key point: Fast, global, guides products: but treat the patient, not the trace.

7. Massive haemorrhage protocol

Key points

- RBC:FFP:platelets ≈ 1:1:1 until results (PROPPR)
- TXA 1 g within 3 h in trauma (CRASH-2) and PPH (WOMAN)
- Permissive hypotension in trauma (not TBI)
- Monitor Ca, K, temperature, lactate; avoid > 2 L crystalloid

Key point: Activate early, balanced products, then switch to goal-directed.